Great Wall: Reinventing Coercion-Resistance

"I have never seen an unworried Bitcoiner"

The 2023-2026 Bitcoin cycle has seen a terrifying rise in **wrench attacks** (theft of crypto through robbery or kidnapping), and the community still has no simple, effective, affordable answer. Yet. Meet **Great Wall**, the definitive coercion-resistant self-custody protocol — **the only** one combining all four properties below:

Key Features

1 Deviceless / KBA

No gadget, metal plate, vault, physical address, etc (additional points of failure) are necessary for the key derivation.

2 Non-Shared Custody

You don't compromise Bitcoin's foundational premise of **individual** custody.

3 Non-Obscurity

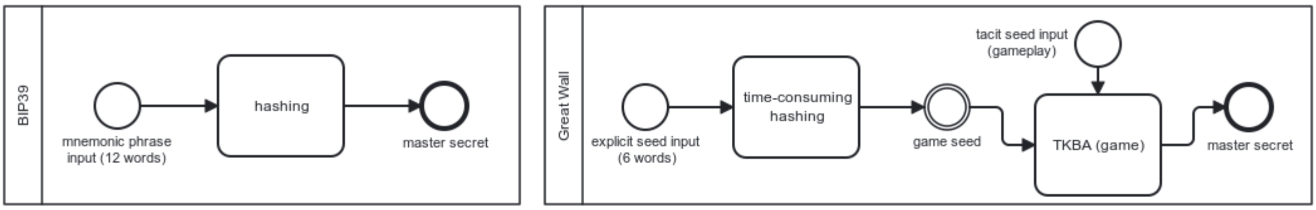
You don't need to hide in shadows, nor wish on your lucky start that your secret trick (like decoy wallet) is never learned by the bad guys.

4 Coercion-Resistance

Commitment or threat of violence on user yields no material benefit to the attacker in any circumstance.

How It Works

Simply put: "1) It's all in your brain; 2) in nobody else's; 3) the attacker knows it; and 4) still cannot coerce it out of you". Here is how we deliver that — and monetize it.



Comparative BPMN diagrams representing BIP39 and Great Wall key derivation schemes.

The protocol is **TGPO — the Time-Gated Perceptual Oracle**. It authenticates by *recognition* rather than recall: you recognise something you cannot describe, so there is nothing to hand over under coercion. In its **self-custodial** deployment, likewise BIP39, it's just a key derivation scheme that can run 100% offline in any device. The simple way to describe it is to say:

1. You input an initial seed equivalent in strength to 6 BIP39 words into an hours-long hash;
2. Hash specifies a seed to your 'private game', which admits a near-infinity of possible 'gameplays';
3. Bits of your 'private gameplay' of your 'private game' are added to the entropy of the master secret;

The catch is that the knowledge of 'gameplay' is easy to memorize but impossible to put into words. Because of that, you, kept alive and well, are strictly necessary during the entire process, which can take hours, days, or even weeks — so stealing your stash becomes at least as difficult as sustaining a kidnapping that long. That fractal game is the **self-custodial** deployment, needing $\sim$**160 bits** on your own device, since nobody counts guesses against an encrypted seed. It also ships **custodially**: the *institution* imposes the delay and counts failed attempts, so $\sim$**16 bits** suffices — four palettes of sixteen server-synthesised **faces**, one pick each: $16^4 = 2^{16}$, against 10^4 for a 4-digit PIN. *(Self-custodial prototype partial; custodial designed, not built.)* Likewise Anki and Duolingo, an **integrated memory coach** prevents both *forgetting* and *fatigue*: a half-minute game-like UX, replayed about weekly, cements lifelong coercion-resistant access.

Monetization — first-mover lock-in

1. **subscriptions** (individuals) — without compromising security, outsource the hours- or days-long computation instead of tying up your own device; **inheritance** rides the same recurrent micropayment;
2. **licensing** (institutions) — custodial TGPO licensed to banks and fintechs, which impose the delay themselves: licence, seats and integration fees, no take rate by construction;
3. **capacity sales** (companies) — solving capacity for *digital* assets: treasury keys, signing credentials, root secrets;
4. **hardware** — for premium UX, high-end users will prefer dedicated air-gapped hardware;
5. **merchandise** — likewise security firms' warning signs, dissuades attackers and propagates virally;

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